# Python Project - Week 3

## Functions - Battle System

### Today's goal

Use functions, parameters, RETURN values, conditions, and loops to add a reusable battle system to the project.

### 1. Guided Coding

Study the example code below with your team. Type it into your editor and run it. Do not copy and paste.

```
def heal(hp, amount):
    hp = hp + amount
    return hp

player_hp = 80

print("HP before healing =", player_hp)

player_hp = heal(player_hp, 20)

print("HP after healing  =", player_hp)
```

### 2. Team Coding Tasks

1. Create a battle(player, enemy_name, enemy_hp, enemy_attack) function.

2. Use a while loop so that the battle continues while both the player and enemy can fight.

3. Let the player choose a to attack or r to run.

4. Return True when the enemy is defeated and False when the player runs away or loses.

5. If battle() returns True, give the player 20 gold and add an item to the inventory.

### 3. Expected Output / Checkpoint

```
A slime appeared!
Your HP: 100 / slime HP: 35
Attack or run? (a/r): a
...
You defeated slime
Reward: 20 gold
Item obtained: slime jelly
```

**4. Submission Check**

• The program runs without errors.

• The program runs from beginning to end.

• Every team member can explain the code added this week.

• Save the completed source code as week3_project.py.